MIL-B-36411(DM)
15 March 1965

MILITARY SPECIFICATION

BLADE AND FRAME, DENTAL LABORATORY DIE SAW

1. SCOPE

1.1 This specification covers a saw frame and blade suitable for cutting dies from casts in crown and bridge procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-up, Paperboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-12902	Packaging of Surgical, Dental and Veterinary Instruments, Diagnostic Equipment, and Related Items.

STANDARDS

Federal

Fed. Test Method Std. No. 151 Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6520

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DRAWING

Defense Medical Supply Center

20899 Post's Improved Die Saw Frame

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Frame. The inverted U shaped portion of the frame shall be formed of SAE 1010 round steel rod and be connected to a wooden handle by a steel ferrule fabricated from 0.015 (No. 28 U. S. Standard Gage), or heavier gage, sheet steel. The wooden handle shall be thoroughly seasoned, straight clear grained, solid hardwood, suitable for the purpose intended.

3.1.2 Blade. The blade shall be fabricated of SAE 1070 or W1-10 steel with SAE 1065 steel pins at each end of the blade.

3.2 Style, design and dimensions.

3.2.1 Frame. The saw frame shall be of Post's design for use in dental crown and bridge die cutting with pin end saw blade specified in 3.2.2 and shall consist essentially of an inverted U shaped bow, a wooden handle and a connecting steel ferrule. The frame shall be constructed so as to act as a spring for applying proper tension to the blade when in position, and for maintaining blade rigidity and alignment during use. The bow shall be tightly fitted into the wooden handle and all parts shall be securely connected together. The frame shall be of the style, design and dimensions shown in Defense Medical Supply Center (DMSC) drawing 20899. The maximum weight of the frame shall be 1-1/2 ounces. Unless otherwise specified, the frame shall be furnished without the blade (see 6.1).

3.2.2 Blade. The saw blade shall be of Post's design, for use with the frame specified in 3.2.1 to form a dental saw for dental crown and bridge die cutting. The blade shall be flat with teeth filed and set in alternate opposite directions in accordance with standard commercial practice and shall conform to the dimensions shown in table I. The design of the blade shall include a securely fitted pin through each end which shall engage and tension the blade into the frame. Unless otherwise specified, the blade shall be furnished without the frame (see 6.1).

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Table I. Dimensions

	(inches)
Overall length	4-1/4 ± 1/16
Pin length	5/32 ± 1/32
Center to center distance between pins	4.000 ± 0.010
Pin diameter	0.046 ± 0.002
Thickness of blade and ends	0.010 ± 0.001
Thickness over teeth	0.014 ± 0.001
Width of blade at end	0.100 ± 0.010
Width of blade over teeth	0.063 ± 0.003
Length of cut	3 (min.)
Teeth per inch	18 ± 1

3.3 Hardness. The blade shall be hardened and tempered and shall have a Rockwell hardness of C50 to C55. The blade pins shall be hardened and shall have a Rockwell hardness of C35 to C45. Hardness shall be tested as specified in 4.4.2.

3.4 Finish. All metal parts of the frame shall be bright nickel plated to render them corrosion-resistant. The wooden handle shall be smoothly finished and lacquered or suitably painted black. The back and ends of the blade shall have squared, dressed edges.

3.5 Identification marking. Each frame shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class in every respect. All parts shall be symmetrically made, smoothly and properly finished, and shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in the following table:

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Chemical Composition	3.1.1 and 3.1.2	4.4.1	*	NA
Hardness	3.3	4.4.2	S-2	**

* Sufficient sample to run analysis for each lot of steel.
**Acceptance number shall be zero (0) for all sample sizes.

4.3 Examination. The blade and frame shall be examined to determine compliance with specification requirements.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition of alloy. Chemical analysis shall be determined in accordance with method 111 or 112 of Standard No. 151.

4.4.2 Hardness. Hardness shall be determined in accordance with method 243 (Rockwell) or method 244 (Diamond Pyramid) of Standard No. 151.

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5. PREPARATION FOR DELIVERY.

5.1 Unit of issue.

5.1.1 Stock No. 6520-055-1518. One (1) frame, as specified, constitutes one unit of issue.

5.1.2 Stock No. 6520-055-1519. Six (6) blades, as specified, constitutes one unit of issue.

5.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

FSN	Unit package	Intermediate package	Exterior container
6520-055-1518	1 unit	12 units	576 units
6520-055-1519	1 unit	12 units	144 units

5.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Stock No. 6520-055-1518.

5.3.1 Packaging and Packing.

5.3.1.1 Level A. Packaging and packing shall conform to Procedure Code No. 35 of MIL-P-12902 with Amendment No. 1 dated 25 August 1961, except as follows: Commercial colors shall be acceptable for paperboard cartons when furnished as specified. In addition, at end of 3.4.8.2, MIL-P-12902, add: 'As an alternate, exterior container shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class weather-resistant. Strapping and closure of each fiberboard box shall be in accordance with appendix of box specification. Waterproofing shall be as specified in 30.4 of PPP-B-636. At the end of 3.4.8.2.1 of MIL-P-12902, add: 'Waterproof barrier bag shall not be required for fiberboard boxes."

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5.3.1.2 Level B. Packaging and packing shall conform to Procedure Code No. 21, MIL-P-12902, with Amendment No. 1 dated 25 August 1961, except that commercial colors shall be acceptable for paperboard cartons when furnished as specified. Method I closure of PPP-B-636 shall be utilized for Level B packs.

5.3.1.3 Level C. The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3.2 Marking.

5.3.2.1 Interior packages. Interior packages shall be marked in accordance with MIL-STD-129. When labels are utilized on cartons, waterproofing shall be required only when applicable cartons are fabricated from water-resistant material. Contract or purchase order number and name of contractor shall be shown. In addition, and as applicable, marking as required by 3.6.1.1 and 3.6.1.2 of MIL-P-12902 shall also be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

5.4 Stock No. 6520-055-1519.

5.4.1 Packaging.

5.4.1.1 Level A.

5.4.1.1.1 Cleaning. Each blade shall be thoroughly cleaned of all shop dirt, grit, or other contamination.

5.4.1.1.2 Preservation. Each blade shall be coated with a suitable rust-preventive compound.

5.4.1.1.3 Unit package. Each unit (6 blades) shall be wrapped in anti-corrosive greaseproof paper and packaged in an envelope of suitable size and design. Envelope shall be adequately secured to prevent accidental opening.

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5.4.1.1.4 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.4.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.4.2 Packing.

5.4.2.1 Level A.

5.4.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.4.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.4.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.4.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.4.2.2 Level B.

5.4.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

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5.4.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4.3 Marking.

5.4.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.4.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Frame, when required (see 3.2.1).
(c) Blade, when required (see 3.2.2).
(d) Selection of applicable levels of packaging, packing and marking (see 5.3 and 5.4).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force. - 03

Preparing activity:
Defense Supply Agency. - DM

Review activities:
Army. - MD
Navy. - MS
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MILITARY SPECIFICATION dated,
BLADE AND FRAME, DENTAL LABORATORY DIE SAW MIL-B-36411(DM), 15 March 1965.

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426

SND PP30 12302

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

DEFENSE MEDICAL SUPPLY CENTER

ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE TM5)

3RD AVENUE AND 29TH STREET

BROOKLYN, N.Y., 11232

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